TEXORA · LECTURE SHEET

Dyeing Machinery

At specialisation level the question is no longer what a jet or a jigger is, but which machine a given fabric, dye class and lot size demands — and what its capacity costs. This sheet treats dyeing machines as production units with cycle arithmetic.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 7 · SPECIALIZATION — WET PROCESSING

1. Classification by transport principle

Every dyeing machine is a solution to one problem: repeated, uniform contact between goods and liquor under a controlled temperature programme. Machines divide by **what moves**:

- **Goods move, liquor nearly still:** winch (beck) — gentle, high LR.
- **Both move:** jet, overflow/soft-flow, airflow — rope transport driven by liquor (or air), the modern knit and polyester workhorses.
- **Goods still, liquor moves:** package, beam, cabinet (hank) machines — the goods are wound or packed and liquor is pumped through them.
- **Goods move in open width through a shallow bath:** jigger — the low-LR woven specialist.
- **Continuous:** pad-dye ranges (pad–dry–cure, pad–steam), where dwell is in a steamer or curing chamber, not a bath.

2. The jet machine — anatomy of the standard production unit

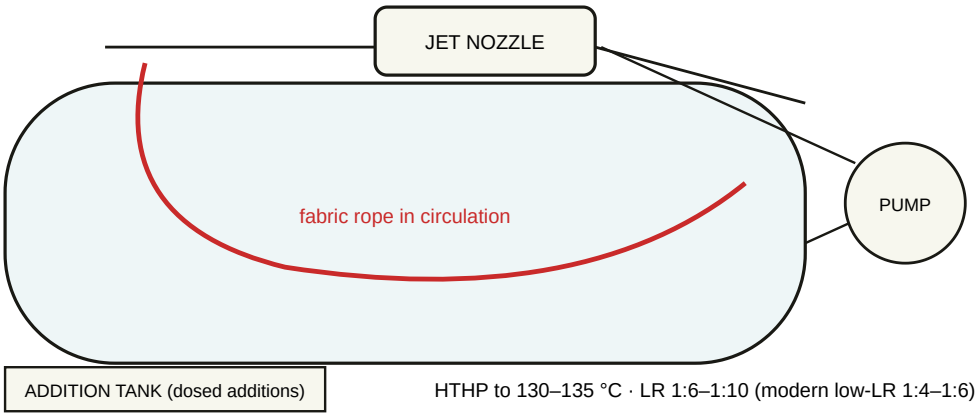


Fig. 1 — Jet dyeing machine (schematic): pumped liquor through the nozzle transports the rope; the addition tank meters chemicals into the circulation so the bath never receives a concentrated shock.

The pump circulates liquor through a venturi nozzle; the jet stream propels the fabric rope around the loop at speeds of a few hundred metres per minute, so every part of the rope meets fresh liquor many times per minute. Sealed HTHP construction reaches 130–135 °C for disperse dyeing of polyester. Fabric is loaded as one rope per **tube (chamber)**; multi-tube machines multiply capacity per batch. Critical operating variables: nozzle pressure and reel speed balanced so the rope circulates without tangling or excessive abrasion (pilling), loading per tube within the maker's range, and programmed dosing from the addition tank (salt and alkali in portions — see WP-4101 Topic 3).

3. The machine park compared

MACHINE	LR	MAX TEMP	GOODS FORM	PRODUCTION CHARACTER
Winch	1:15–1:25	98 °C	Rope, loose in bath	Gentle; delicate goods; high water/salt cost
Jigger	1:3–1:5	98 °C (HT 130 °C)	Open width woven	Lowest batch LR; ending/listing risk; crease-free
Jet (HTHP)	1:6–1:10	135 °C	Rope, knits and wovens	Fast, versatile; the polyester and knit standard
Overflow / soft-flow	1:8–1:15	98–130 °C	Rope, delicate goods	Gentler transport than jet; crease-sensitive fabrics
Airflow	1:3–1:5	130 °C	Rope moved by air	Lowest water/chemical use (Topic 11); higher capital

Package	1:8–1:12	135 °C	Yarn on perforated cones	Precision yarn dyeing; winding quality decides levelness
Beam	1:8–1:12	130 °C	Open width on perforated beam	No rope marks; delicate wovens
Pad range	pick-up based	Cure to 180 °C	Open width, continuous	Highest throughput for woven volume (Topic 6 arithmetic)

4. Capacity arithmetic — the batch machine as a production unit

WORKED EXAMPLE 1 — DAILY OUTPUT OF A 4-TUBE JET

Tube capacity 250 kg → batch = 4 × 250 = **1,000 kg**.

Cycle: load and heat 0.5 h + dyeing programme 4.5 h + wash-off and cool 1.0 h + unload 0.5 h = **6.5 h**.

Working day 20 h (two shifts with hand-over losses) → ⌊20/6.5⌋ = **3 batches** → daily output = 3 × 1,000 = **3,000 kg/day**.

The floor function is the lesson: a cycle of 6.5 h and a cycle of 6.6 h give the same 3 batches, but shortening the cycle to 5.0 h would give 4 batches — a 33% capacity gain with no new machine. Cycle-time reduction (faster heating, rapid wash-off, right-first-time dyeing) is capacity engineering. This arithmetic is the seed of the project plan in Topic 12.

Utilisation must also respect shade and quality changeovers: a machine scheduled across many small lots loses time to cleaning and re-filling. Production planning therefore groups lots by shade family and fibre, and the planner quotes capacity in *effective* batches — typically 80–90% of the arithmetic maximum.

5. Process control points on modern machines

- **Programmed temperature profile** — ramp rates (e.g. 1–2 °C/min through the critical exhaustion zone) are stored per dye class; the profile is part of the recipe (WP-4101 Topic 1 explains why: exhaustion rate follows temperature).
- **Metered dosing** — salt, alkali and dye additions by pump from the addition tank, linear or progressive curves, preventing the unlevelness of hand additions.
- **Load and LR verification** — the machine fills to the volume the booked LR demands for the actual load weighed in; an error here invalidates every g/L figure in the recipe (Topic 2).
- **Rinsing efficiency** — overflow vs batch rinsing programmes trade water against wash-off quality; hydrolysed reactive dye must be fully soaped off or wash fastness fails regardless of dyeing quality.

6. Glossary — key terms

- **Tube/chamber:** one rope circuit of a multi-rope jet machine, with its own loading limit.
- **Cycle time:** total machine occupancy per batch, load to unload.
- **Right-first-time (RFT):** production meeting shade and quality without correction additions or re-dyeing.
- **Dosing curve:** the programmed rate at which a chemical is metered into the bath.

7. Key points

- Select machines by transport principle, goods form, LR and temperature ceiling — then prove capacity by cycle arithmetic, not brochure rates.
- Cycle time, not batch size alone, sets daily output; the floor of (working hours ÷ cycle) governs everything.
- Modern machine value lies in programmable profiles and metered dosing — the machinery of right-first-time dyeing.

8. Summary

Dyeing machinery at this level is production engineering: the jet of Fig. 1 is a 3,000 kg/day unit whose output the planner can raise by a third by shaving 1.5 hours from its cycle. Machine selection and cycle arithmetic together decide whether a dyehouse can promise a delivery date — and keep it.

9. Practice

1. A 2-tube jet loads 300 kg per tube. Its cycle is 5.5 h and the working day is 21 h. Calculate batches per day and daily output. What cycle time would be needed for 4 batches?
2. Choose machines, with reasons, for: (a) silk-look delicate polyester crepe that creases easily; (b) 2,000 kg/day of cotton single-jersey with reactive dyes; (c) yarn for striped knitted fabrics requiring several colours.
3. Why does an error in booked load weight invalidate the g/L parts of a recipe even if the machine fills correctly to its programmed volume?

Answer key

1. Batch = 600 kg. Batches = $\lfloor 21/5.5 \rfloor = 3$ → output 1,800 kg/day. For 4 batches the cycle must be ≤ 21/4 = 5.25 h.

2. (a) Overflow/soft-flow — gentle flooded transport, low creasing. (b) Jets sized by the Topic-12 arithmetic (e.g. 1,000 kg-batch machines, 3 batches/day class) — rope form suits jersey, HTHP not needed for reactive but the circulation gives levelness. (c) Package dyeing — yarn dyed before knitting so stripes are knitted from coloured yarns.
3. g/L figures are computed against bath volume = load × LR. If the load is wrong, the machine fills the volume for the booked (wrong) LR, so every bath-dosed chemical's concentration — and its % owf equivalent — shifts by the load error, while % owf dye (weighed on the true goods mass) stays right: the recipe's internal balance breaks.

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